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1 using Handcalcs, PlutoUI, LaTeXStrings, PlutoPlotly;
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Definitions

Electrostatic discharge

Parasitic parameters on PCB cause high voltage, current.

$$C'_{par} = 5 \frac{pF}{cm}$$

$$L'_{par} = 10 \frac{nH}{cm}$$

$$U = L_{par} \frac{di}{dt} = 10nH \cdot \frac{10A}{1ns} = 100V$$

$$I = C_{par} \frac{du}{dt} = 5pF \cdot \frac{3kV}{1ns} = 15A$$

Switching inductive loads

When opening a switch, relay connected to an inductive load, voltage will be induced through its contacts. Voltage could be several kV, causing a light arc.

Coupling mechanism

Common mode

Common mode

- U, I flows one way
 - ground is reference potential for all voltages
- $u_{cm} = \frac{u_1 + u_2}{2}$
- $i_{cm} = \frac{u_{cm}}{Z_{cm}}$
- prone to EMI because of synchronous clock

Differential mode

Differential mode

- U, I flows both ways
 - signal gets inverted
 - looking at voltage difference
- $u_{dm} = u_1 - u_2$

Coupling between current loops

Coupling between U_q with i_{used} , Z_1 and U_{noise} with i_{noise} , Z_2

- Impedance coupling
 - caused by common ground
- capacitive coupling
 - caused by $Z_2 \rightarrow \infty$
- inductive coupling
 - caused by $Z_2 \rightarrow 0$

Coupling on PCB

- coupling impedance of ground: $Z_O = R_O + j\omega L_O$
- minimizing coupling through small current loops, high distance between tracks

Capacitive coupling

- Assumption: $Z_{C,12} \ll Z_2 \rightarrow i_{C,12} \propto i_{noise} = \frac{u_{noise}}{Z_{C,12} + R_i || Z_1}$
- HF-noise on load : $u_{L,noise} = i_{noise} \cdot R_i || Z_1 = \frac{u_{noise} \cdot R_i || Z_1}{Z_{C,12} + R_i || Z_1}$
- Countermeasures
 - low-pass-filter
 - reduced capacity through
 - reduced length of coupled tracks
 - higher distance between tracks with different current loops
 - avoid parallel tracks
 - electric shielding
 - ground vias for magnetic field catching

Filter

Introduction

- Goals
 - reduce emission
 - compatibility
- no filter optimal for all requirements

Topology

- linear filter
 - combination of R, L, C
 - low-pass
 - high-pass
 - band-pass
 - band-stop
- non-linear filter
 - diodes, thyristors, triacs, varistors, gas dischargers

Low-pass first order

- Transfer function: $H(j\omega) = \frac{U_{out}}{U_{in}} = \frac{1}{1+j\omega RC}$
- Amplitude frequency response: $|H(j\omega)| = \frac{1}{\sqrt{1+\omega^2 R^2 C^2}}$
- Phase frequency response: $\arg(H(j\omega)) = -\arctan(\omega RC)$
- Cutoff frequency: $f_g = \frac{1}{2\pi RC}$

Low-pass second order

- Transfer function: $H(j\omega) = \frac{U_{out}}{U_{in}} = \frac{1}{1-\omega^2 LC + j\omega RC}$
- Amplitude frequency response: $|H(j\omega)| = \frac{1}{\omega C \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}}$
- Phase frequency response: $\arg(H(j\omega)) = -\arctan(\frac{\omega RC}{1-\omega^2 LC})$
- resonance frequency: $f_r = \frac{1}{2\pi\sqrt{LC}}$
- absolute bandwidth: $b_\omega = \frac{R}{L}$
- relative bandwidth: $b_{rel} = \frac{b_\omega}{\omega_r} = R \cdot \sqrt{\frac{C}{L}}$
- quality: $Q = \frac{1}{b_{rel}} = \frac{1}{R} \cdot \sqrt{\frac{L}{C}}$

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Shielding

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Casing connector

- ensure large, conductive connection
 - conductive foam
 - foam
 - enclosed by nickel-copper-sinew
 - contact spring trap
 - copper-beryllium/stainless steel

Base material	Nickel-Copper	Aluminium
Zinc	-	++
Aluminium	-	++
Copper	+	-
Zinn	+	-
Nickel-Silver	+	-
Lead	+	-
Nickel	++	-
Silver	++	-
Nickel-Copper	++	-
Gold	++	-

Wiring

- conductive wiring harness, cable clamps

PCB

- ground plane as electric shielding
- noisy components should be shielded locally

Ground connection

- conductive casing part, ground planes should have low-impedance connections
- mechanical variants
 - standoff bolt
 - grounding cable
- conductive surface coated contact springs
 - copper-beryllium
 - phosphor-bronze

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1 md"""### Ground connection
2 - conductive casing part, ground planes should have low-impedance connections
3 - mechanical variants
4   - standoff bolt
5   - grounding cable
6 - conductive surface coated contact springs
7   - copper-beryllium
8   - phosphor-bronze
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